Temporal ordering of cognitive impairment in Parkinson's disease based on disease progression models

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A B S T R A C T

Introduction: Identifying Parkinson's disease (PD) patients at risk of cognitive decline is crucial for enhancing clinical interventions. While several models predicting cognitive decline in PD exist, a new machine learning framework called disease progression models (DPMs) offers a data-driven approach to understanding disease evolution.

Methods: We enrolled 423 PD patients and 196 healthy controls from the Parkinson's Progression Markers Initiative (PPMI). Our study encompassed a range of biomarkers, including motor, neurocognitive, and neuroimaging evaluations at baseline and annually. A methodology was employed to select optimal combinations of biomarkers for constructing DPMs with superior predictive capabilities for both diagnosis and estimating conversion times toward cognitive decline.

Results: At baseline, the approach demonstrated high performance in detecting subjects at high risk of converting into cognitive decline in the first five years. Furthermore, the proposed timeline from cognitive unimpaired to dementia was also used to explore clinical events such as the onset of cognitive impairment, the development of dementia or amyloid pathology. Presence of amyloid pathology did not alter progression in cognitive impairment among PD subjects.

Conclusions: Neuropsychological measures and certain biomarkers, including cerebrospinal fluid (CSF) $A\beta_{42}$ and DAT deficits, predict cognitive decline and estimating a timeline from cognitive unimpaired to dementia, with amyloid pathology preceding the onset of dementia in many cases. Our DPMs suggested that the profiles of CSF Aβ_{42} and pTau in PD may differ from those in aging and Alzheimer's Disease.

Keywords:

Parkinson's disease

Cognitive decline

AD CSF Biomarkers

Disease progression models

1. INTRODUCTION

Parkinson's disease (PD) is a common neurodegenerative disease identified by motor symptoms such as tremor, rigidity, and bradykinesia. Additionally, PD is associated with non-motor symptoms like depression, anxiety, and cognitive impairment (1) (2), which significantly raise the risk of dementia (3). Identifying individuals with PD at risk of cognitive impairment could help stratify PD cohorts for clinical trials, to implement appropriate interventions, to optimize medical management, and to enhance patients' autonomy.

The progression of cognitive decline in PD seems to follow a continuum from cognitive unimpaired (PP-CU) to mild cognitive impairment (PD-MCI) and, in later stages, PD dementia (PDD). PD patients have a significantly higher risk of developing dementia compared to healthy individuals (4) (5). However, the timeline for individual patients varies greatly, and some PD patients may remain cognitively unimpaired for an extended period after diagnosis (6).

Several risk factors associated with cognitive decline in PD have been identified (7). These factors include age, olfactory dysfunction (8), rapid eye movement (REM) sleep behavior disorder (9), severity of motor symptoms (10), mood variables (11), and cerebrospinal fluid (CSF) biomarkers (12). Regarding CSF risk factors, pathological biomarkers of Alzheimer's disease (AD), such as amyloid-β (Aβ) plaques and tau neurofibrillary tangles, are found to coexist with α-synuclein aggregates in PD, and their accumulation is believed to impact on cognitive impairment along PD progression (13). However, the role of Aβ in PD-related cognition is complex and not well understood (14) (15) (16) (17).

Several different predictive models for predicting cognitive decline and risk conversion into PD-MCI or PDD have been reported using either baseline or longitudinal data (18) (19) (20) (21) (22) (23) (24). In this regard a new machine learning framework which contemplates disease progression as a continuum has been proposed (25) (26) (27)(28)(29). The framework, known as disease progression models (DPMs), provides a data-driven description of the natural evolution and automatic diagnosis of diseases by explicitly ordering biomarkers from normal to pathological stages along the disease time axis in a multivariate manner. One class of DPMs, such as the growth models by alternating conditional expectation (GRACE), models long-term pathological trajectories from short-term clinical data using self-modeling regression (25).

In this study, we developed DPMs for cognitive decline in PD using data from the Parkinson's Progression Markers Initiative (PPMI) database. DPMs were developed based on quality metrics for the estimation of diagnosis and the prediction of convert times towards cognitive impairment. Simultaneously, proposed natural histories regarding the progression of cognitive decline were utilized to estimate the time remaining until dementia onset or to perform analyses on the influence of amyloid pathology in cognitive decline.

2 Materials

2.1 Subjects

Data were obtained from the PPMI initiative (www.ppmi-info.org/data) including a cohort of 423 drug-naive PD patients and 196 healthy control subjects (HC). The PPMI is a longitudinal multisite study of newly diagnosed (two years or less) and untreated PD patients (30). Inclusion criteria for PD patients within PPMI were recent idiopathic PD diagnosis, being untreated for PD, having a dopamine transporter (DAT) deficit on imaging, and absence of dementia at baseline visit as determined by the site investigator(30). Clinical data was collected from patients for up to 5 years after the baseline assessment.

2.2 Clinical and biomarkers assessment

Participants underwent comprehensive assessments, including motor, neurocognitive, and neuroimaging evaluations at baseline and annually. Relevant demographic, clinical, neuroimaging, and CSF biomarkers were collected (31) (19) (23): Movement Disorders Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS) Part-III scores, Montreal Cognitive Assessment (MoCA), University of Pennsylvania Smell Identification Test (UPSIT), Geriatric Depression Scale (GDS), REM Sleep Behavior Disorder Questionnaire (RBDSQ) and SCOPA-Autonomic (SCOPA-AUT). The following neurocognitive measures (NMs) were also collected: Hopkins Verbal Learning Test Revised (HVLT-R), Benton Judgment of Line Orientation 15-item version (JLO), Symbol-Digit Modalities Test (SDMT), Letter-Number Sequencing (LNS) and the semantic fluency test (SFT). Striatal binding ratios from DAT SPECT scans (MEANCAU, MEANPUT) and CSF analysis metrics (ASYN, Aβ42, tTau, pTau) were also obtained (32) (33).

The cognitive status of PD patients was categorized as PD-CU, PD-MCI, or PDD based on clinical diagnoses, according to site investigator's annual clinical diagnosis, resulting in groups of 323 PD-CU, 83 PD-MCI, and 17 PDD subjects Assignments to these three clinical groups were determined based on the last diagnosis of each patient.

2.3 Statistical Analysis

Since the natural history of PD cognitive decline estimated by GRACE depends on the choice of selected markers, a predictive modeling approach was employed to identify subsets of markers from longitudinal data that exhibited strong performance in detecting a high risk of cognitive impairment progression (34) (see supplementary materials, S1). GRACE incorporated subject-specific time shifts, fixed effects, random intercept and slope, and measurement error terms, estimating time shift parameters and short-term and long-term curves of selected markers (25). The training of the DPM did not involve follow-up clinical diagnostic information. Bootstrapping techniques were applied to obtain confidence intervals for DPM evaluation parameters. They were also used to analyze the long-term trajectories of markers in patient groups with and without amyloid pathology.

As mentioned, GRACE estimates the time shift of each i-th patient in the progression of cognitive decline, denoted as δ_i. This scalar value transfers the marker trajectories of visit times to the proposed time frame for the progression of cognitive decline. It also models the short-term trajectories of each marker and patient using the centered time of the visits (t_{ij}^c). Specifically, $t_{ij}^c = t_{ij} - \left(t_{i_1} + t_{i_{end}}\right)/2$, where t_{ij} is the j-th visit time, and i_1 and i_{end} indicate the first and last visit index of the i-th subject. For the initial year of cognitive impairment or zero time, represented as t_{onset}, the trajectories of PD-CU subjects are expected to be to the left of t_{onset}, while PD-MCI/PDD subjects' trajectories should cross t_{onset} towards the right. Therefore, when estimating temporal ordering using GRACE with a proposed subset of markers, certain fitness scores can be proposed. The first score we defined is the percentage of PD-CU subjects whose last visits, $t_{i_{end}}^c + \delta_i$, were less than t_{onset} in relation to the total number of PD-CU subjects:

$$SCORE_1 = \frac{\#\left\{i|(i \in PD-CU) \cap \left(\left(t_{i_{end}}^c + \delta_i\right) < t_{onset}\right)\right\}}{\#PD-CU}$$

Regarding PD-MCI/PDD subjects, two measures were established: a) The proportion of PD-MCI/PDD subjects who did not have cognitive decline at baseline, $\left(t_{i_1}^c + \delta_i\right) < t_{onset}$, compared to the total number of PD-MCI/PDD subjects:

$$SCORE_2 = \frac{\#\left\{i|(i \in PD-MCI/PDD) \cap \left(\left(t_{i_1}^c + \delta_i\right) < t_{onset}\right)\right\}}{\#PD-MCI/PDD}$$

and b) the ratio of PD-MCI/PDD subjects whose last visit was greater than t_{onset} with respect to the total number of PD-MCI/PDD subjects:

$$SCORE_3 = \frac{\#\left\{i|(i \in PD-MCI/PDD) \cap \left(\left(t_{i_{end}}^c + \delta_i\right) > t_{onset}\right)\right\}}{\#PD-MCI/PDD}$$

Hence, a zero time was estimated by maximizing the three ranking scores mentioned earlier. As t_{onset} is increased, $SCORE_1$ and $SCORE_2$ also increase, while $SCORE_3$ decreases. The optimal time is the one that achieves the highest score for all the three measures. Further, it is also preferable for these values to be similar to each other, therefore we establish the following criteria to minimize the absolute differences in the $SCORE_i$ values among the PD subjects:

$$\hat{t}_{onset} = \arg\min_{t_{onset}} \sum_{r=1}^{2} \sum_{s=r+1}^{3} |SCORE_r(t_{onset}) - SCORE_s(t_{onset})|$$

For each vector used to train GRACE, a timeline of the disease is proposed. To select the most likely natural history, three criteria are proposed to measure the reliability of the suggested DPM: i) Consider the three scores from the previous subsection, $SCORE_i$. ii) Perform a linear regression analysis with the PD-MCI/PDD subjects, comparing their conversion times to the times estimated by the proposed DPM, $t_{onset} - \left(t_{i_1}^{c} + \delta_i\right)$, for all i ∈ PD-MCI/PDD. And iii) Compare the values of the long-term trajectories of the markers at t_{onset} and their cutoffs used in clinical practice to diagnose patients into cognitive impairment. Therefore, among the vectors that showed good performance in predicting a high risk of progressing into cognitive decline at baseline, these three criteria were used to select the optimal combination of markers to train GRACE.

3. Results

3.1 Cross sectional analysis

At baseline, the scores of specific cognitive domain tests, including LNS, SDMT, JLO, IMMEDIATE, HVLTRDLY, SFT, and SCOPA, as well as the clinical variables UPSIT and RBDSQ, and the biomarkers pTau/Aβ_{42} and MEANCAU, showed significant differences between the PD-CU subjects and the PD-MCI/PDD patients (*see Table 1*). Note that these clinical groups were determined in the last available follow-up of the patients. No significant differences between groups were found for MOCA. It is worth observing that the patients in the PD-MCI/PDD group (mean age 66.0 and 69.4 years) were older than those in the PD-CU group (mean age 60.1 years). The comparison between HC and PD patients revealed that most of the NMs showed significant differences, indicating a poorer cognitive state in the PD group despite age matching. Regarding AD CSF biomarkers, tTau and pTau were lower in the PD group in relation to HC patients, and no differences were also observed for both in Aβ_{42} and in pTau/Aβ_{42} (see supplementary materials, S2).

[Table 1]

3.2 AT profiles in PD

The AT profile is a well-established framework used to assign stages of AD based on biomarkers of amyloid (A) and tau (T) (35). In this framework the cut-off for amyloid pathology (A+) was established as $A\beta_{42} \leq 683.45 pg/ml$ (15), for tau pathology (T+) as $pTau > 23 pg/ml$ (36) (37). Of the entire cohort, 374 of 423 PD patients and 174 of 196 healthy controls (HC) had CSF biomarker data available. At baseline, 23.4% of the PD-CU, 26.5% of the PD-MCI, 66.7% of the PDD and 22.4% of the HC were A+. There was no significant difference between the PD-CU vs PD-MCI, or between HC-CU vs HC-MCI subgroups in relation to amyloid pathology. No significant differences were also found between the PD cohort and HC in amyloid pathology at baseline. However, there were significant differences with respect to the subjects who developed PDD. At 5-years follow-up, the percentages of all clinical groups with A+ increased, with 35.7% of the PD-CU, 39.7% of the PD-MCI, 66.7% of the PDD and 32.2% of the HC. Still, no significant differences were found between PD vs HC, nor between the clinical groups of PD-CU vs PD-MCI, or between HC-CU vs HC-MCI. Again, the percentage of A+ in the PDD group was significantly higher than in the other groups. Tau pathology incidence was less frequent than amyloid. Isolated tau pathology was below 10% in all groups. Only the patients that developed PDD had an incidence of 20.7% for isolated tau pathology (A-T+) and an incidence of 26.7% for combined amyloid and tau pathology (A+T+), indicating coexistence with AD.

The measures pTau and tTau were highly correlated (R=0.98-0.96) at baseline in all clinical groups. Correlations between α-Syn and AD CSF biomarkers ($A\beta_{42}$, pTau) were

moderate (R=0.59-0.88). No correlation was found between the AD CSF biomarkers and NMs (see supplementary materials, S3).

3.3 Predictive models

The best predictive model of progression from the PD-CU stage to PD-MCI/PDD proposed the following vector of features:

$$\{\text{HVLTRDLY,LNS,SDMT,SFT,A}\beta_{42}\text{,MEANCAU}\},$$

among the other proposed vectors (see supplementary material for details, S4). At baseline, the predictive model showed a good classification between PD-CU versus PD-MCI/PDD patients. The sensitivity (SEN) was 73.7% (70.9%-76.6%), specificity (SPE) was 73.5% (71.9%-75.1%), accuracy (ACC) was 73.4% (72.0%-74.7%), and area under the curve (AUC) was 0.796 (0.779-0.821) (38).

3.4 DPM

With the studied population, the proposed vector was used for the training of the GRACE approach. Fig.1 shows both the number of the subjects as a function of the time shift, δ_i, or "cognitive decline times" as well as the relationship between the conversion times of the PD-MCI or PDD subjects, $t_{convert}$, and their estimates by GRACE, $\hat{t}_{convert}$.

[Fig.1]

The regression line was $\hat{t}_{convert} = 2.12 \cdot t_{convert} - 7.36$ and with a Pearson correlation coefficient of $R = 0.55\ (0.53 - 0.56)$, while using patients' age was $\widehat{age}_{convert} = 0.74 \cdot age_{convert} - 12.75$ and with $R = 0.76\ (0.74 - 0.77)$. By optimizing the classification scores based on $t_i^c + \delta_i$ and the clinical groups, the cognitive decline onset was estimated to $\hat{t}_{onset} \approx 5.2$ years. For this $\hat{t}_{onset}$, $SCORE_1$ = 82.3% (81.6% – 82.9%), $SCORE_2$ = 62.4% (60.1% – 64.7%) and $SCORE_3$ = 66.5%(64.4%–68.6%).

Fig. 2 shows the individual observed trajectories for each feature, in the original scaling, ordered in the progression timeline using the DPM approach. The long-term trajectories were also highlighted in each feature subplot. Note that the GRACE approach is blind to the information of the clinical category. Even so, the short-term trajectories were painted with different colors depending on the clinical group of the subjects at the last follow-up (PD-CU, PD-MCI, or PDD). This visualization shows that the proposed ordering was consistent with the clinical classification.

[Fig.2]

Once the progression timeline from PD-CU to PD-MCI or PDD was defined from the δ_i values of the subjects in the studied cohort, it is possible to infer long-term trajectories for any acquired features, even those that were not used to train the GRACE algorithm. Some events were estimated in the progression into the dementia from the long-term trajectories of Aβ_{42}, pTau or MOCA: a) $\hat{t}_{A\beta_{42}=683\text{pg/ml}} = 6.8\ years$, b) $\hat{t}_{pTau=23\text{pg/ml}} > 15\ years$, c) $\hat{t}_{MOCA=25} = 5.7\ years$ and c) $\hat{t}_{MOCA=21} = 13.3\ years$. These times suggested that the MOCA=25 time and the estimated onset time were similar (5.7 years vs 5.2 years). Dementia would be reached at 13.3 years (MOCA=21). Amyloid pathology would develop after the onset of cognitive decline and before dementia (6.8 years). It can also be inferred that there is an estimated time period of 7.6 years between the development of cognitive decline (MOCA=25) and dementia (MOCA=21). For transforming these times to the PD temporal evolution, the regression line of the converters could be used.

Fig.3 shows the application of bootstrap techniques on the long-term trajectories of some features between the PD individuals classified as A+ or A- at baseline. The results show that there was no acceleration of cognitive decline among the subjects who had amyloid pathology at baseline compared to those who did not.

[Fig.3]

4. Discussions

A data-driven approach was proposed for predicting conversion to PD-MCI and dementia and for estimating the timeline of cognitive decline progression. This approach combined supervised and unsupervised learning techniques. A reduced vector of markers was selected to train the models, which is easily interpretable and generates robust, verifiable, and reliable models. At baseline, the approach demonstrated high performance in detecting subjects at high risk of converting from PD-CU to PD-MCI/PDD in the first five years. Furthermore, the proposed timeline from cognitive unimpaired to dementia was also validated through the diagnosis of patients and the observation of clinical events such as the onset of cognitive decline, dementia development, or amyloid pathology.

In our study the conversions from PD-CU into PD-MCI/PDD subjects within the five years of follow-up was of 24%. This result is consistent with other studies using the PPMI database (39) (22). In other cohorts this incidence was of 43.4% at 3.2 years (40) or 38% at 3 years (41). This variability in conversion ratios could be due to the lack of consensus guidelines among the different cohorts. The introduction of the MDS criteria reduced the heterogeneity in the reported epidemiology of PD-MCI, but there is still variability (42). On the other hand, the control population (HC) in the PPMI study had a conversion rate to MCI of less than 3% after 5 years. Moreover, 25.7% and 22.4% of the PD and HC patients were A+ at baseline, respectively, and there were no significant differences between the two groups. Additionally, after 5 years, 29.2% of the PD-A+ patients converted to PD-MCI/PDD, while the conversion rate for PD- A- patients was 19.8%. These results showed no significant differences ($p=0.056$). In a study conducted by Fiorenzato et al. (43), it was found that 21% of PD patients already had positive cerebral amyloid [18F]florbetaben uptake at the beginning of the disease. However, the overall prevalence was in line with the literature's age-matched control population, suggesting that PD does not necessarily increase the risk of amyloid deposition.

There is controversy surrounding the interpretation of AD CSF biomarkers in the cognitive decline of PD subjects. Some authors associate low CSF $A\beta_{42}$ values with the development of MCI or dementia (14) (15), while others find no evidence of an association between amyloid pathology and cognitive impairment (16) (44). In a recent study, tau pathology was also not associated with MCI in PD (45). In this study, CSF $A\beta_{42}$ levels did not show significant differences between the PD population and the HC group (see supplementary material, S2), or between PD-CU and PD-MCI, as highlighted by other authors (43). We also observed that there was no greater cognitive decline among PD-A+ subjects relative to PD-A- subjects when applying the bootstrap technique in temporal modeling of progression in cognitive decline in PD-novo patients (see fig. 3). It is notable that the tTau and pTau values of the PD subjects were low compared to the control subjects at baseline (see

supplementary material, S2-S3 and (12)), in the opposite direction to what happens in AD (12) (35). Furthermore, there were no significant differences between the CSF amyloid pathology biomarkers in the comparative PD vs HC, both in $A\beta_{42}$ and in pTau/$A\beta_{42}$. On the other hand, the PD-MCI population showed very low percentages of A+T+ profiles, and even in the PDD population. It is lower in PD compared with 60-80% in the case of dementia (46) (47). All of these findings suggest that the profiles of CSF $A\beta_{42}$ and pTau in PD may differ from those in aging and AD (12) (15).

$A\beta_{42}$ Recent evidence suggests that individuals with prodromal features of PD, such as hyposmia, REM sleep behavior disorder, and reduced dopamine transporter binding, may exhibit worse cognitive performance compared to people without any or with only one of these features (49). Older age appears to be the most consistent demographic risk factor in PD-novo (22), whereas the contribution of male gender and fewer years of education is less consistent (40) (23). These findings are consistent with the results reported here, where patients who progressed towards cognitive decline were older, exhibited olfactory disorders, and had lower cognitive scores at baseline.

A single marker, at baseline, was not enough to estimate whether the PD-novo patient will progress towards cognitive decline in the coming years. For example, MOCA obtained, after correction for age, a sensitivity of 87% and a specificity of 48%, in line with other authors (50). Several authors had proposed multivariate predictive models to estimate conversion to cognitive decline at baseline. A neuropsychological measure battery was required, as cognitive decline mostly affects multiple domains rather than a single domain (18). For example, patients who scored lower on working memory and processing speed were more likely to develop cognitive decline (22). Most of the proposed predictive models for cognitive decline in PD have utilized a diverse array of both neuropsychological and biological or imaging features. These models often incorporate variables like DAT imaging, AD CSF biomarkers, UPSIT, REM sleep behavior, MDS-UPDRS, and mood assessments. Our proposed set of features with optimal predictive accuracy included the Hopkins Verbal Learning Test-Revised Delayed Recall, Letter-Number Sequencing, Symbol Digit Modalities Test, Semantic Fluency Test, $A\beta_{42}$, and mean caudate uptake from DAT-SPECT images. These findings suggest that lower CSF $A\beta_{42}$ levels, DAT deficits in the caudate, and certain neuropsychological test scores are indicative of potential cognitive decline in PD. This model aligns with others that have been proposed (19) (20) (23) (39), showing comparable accuracy.

A DPM was selected for its ability to assign the patient diagnosis utilizing the three proposed scores, as well as for its validity in estimating the conversion times towards cognitive decline in PD patients. In this regard, our DPM revealed a long-term trajectory of MOCA that aligns with the proposed natural history of cognitive decline as the time to reach MOCA=25 was similar to t_{onset}. In addition, DPMs can be used to validate hypotheses about disease progression, aligning clinical events with

their estimates. According to our DPM, dementia manifests 7 years after cognitive decline, and amyloid pathology appears approximately 3.5 years later. However, the problem of adapting the estimated times by GRACE with the true natural history of cognitive decline in PD-novo subjects remains open. The regression line between PD-novo subjects progressing to PD-MCI or PDD could be used. However, the Pearson coefficient values were not high enough to give reliable conversions between GRACE and the cognitive timeline among PD-novo subjects. Beyond the timelines, a chronological order of clinical events was suggested. This is consistent with observations indicating that amyloid pathology is present 5 to 10 years before the onset of AD (51) and, 67% of the PDD subjects of the studied cohort showed A+ at baseline before progressing into dementia. Another result of the DPMs is their ability to demonstrate the effectiveness of markers in capturing cognitive decline. In Figure 2, the short-term trajectories of the patients' neuro-psychological measures (HVLTRDLY, LNS, SDMT, SFT) show less dispersion around their long-term trajectories. In contrast, the CSF $A\beta_{42}$ biomarker exhibits greater short-term variability compared to its long-term trajectory, indicating that neuro-psychological measures more accurately capture the progression of cognitive decline.

The long-term trajectories of some selected markers on percentiles showed that NMs exhibit similar linear behavior over the timeline (see supplement material, S5). On the contrary, CSF $A\beta_{42}$, pTau, and MEANCAU deviated from the NMs. CSF biomarker pTau did not progress in any direction in the evolution of cognitive decline. CSF $A\beta_{42}$ biomarker showed already an alteration above 25% more than 15 years before the onset of cognitive decline. Moreover, MEANCAU had the steepest temporal slope in the evolution towards cognitive decline. In the studied cohort, 40% and 67% of PD-novo subjects who progressed to PD-MCI and PDD, respectively, had amyloid pathology after 5 years of follow-up. In contrast, pTau values were lower in PD-novo than in HC subjects at baseline. Furthermore, less than 10% and 30% of subjects who progressed to PD-MCI and PDD, respectively, in the longitudinal analysis reached tau pathology. These conclusions were also supported by observing the long-term trajectories of $A\beta_{42}$ and pTau in percentages. While $A\beta_{42}$ was observed to evolve with cognitive decline, this trend did not appear in pTau.

For the sake of completeness, other DPMs were also built with different combinations of features obtained from our predictive models (see supplementary material, S6). Nevertheless, the results of the different temporal orderings proposed by each DPM were quite alike. The onset times of cognitive decline, the Pearson coefficients, and the proposed quality scores were almost identical among the proposed models (see supplementary material, S6). In addition, the selected marker values at the onset times of their long-term trajectories were also coincided, thus reinforcing the validity of our findings.

There are several challenges that need further analysis. Some clinical events were estimated according to the proposed timelines. However, it is necessary to transfer this time axis to the natural history of cognitive decline in PD-novo subjects. The moderate correlation between the conversion times of the PD-MCI/PDD subjects

with the estimates of the proposed DPM suggests that the model's performance needs further improvement. With an increase in the number of patients and years of follow-up, a more reliable natural history could be inferred. Additionally, this methodology should be extended to other cohorts of PD populations with analysis of cognitive decline.

Authors roles

Carlos Platero: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Writing - original draft. **José Angel Pineda**: Writing - review & editing.

Disclosure statement

All authors declare no conflict of interest.

Declaration of competing interest

The authors disclose no actual or potential conflicts of interest.

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ChatGPT, version 3.5, a language model developed by OpenAI (https://www.openai.com) was used for language refinement of this manuscript.

Supplementary material

Below is the link to the electronic supplementary material.

Table 1: Baseline demographic and clinical characteristics of the patients with PD-CU, PD-MCI and PD-dementia using the site investigator's clinical diagnosis. Data are mean (SD) (minimum, maximum) or n (%). ANOVA with Bonferroni post hoc test is used for baseline age, education, neuropsychological scores and CSF and DAT imaging biomarkers, except for gender and APOE4 allele presence where the chi-square test is used. Statistical significance is considered with $p - value \leq 0.01$. (a) Significant compared PD-CU to PD-MCI, (b) PD-CU vs PDD, (c) PD-MCI vs PDD.

Demographic and clinical characteristics					
Subjects	323 (PD-CU)	83 (PD-MCI)	17 (PDD)	p-valuse	Post-Hoc
Female	120 (37.2%)	20 (24.1%)	6 (35.3%)	0.083	-
APOE4	78 (n=295 26.4%)	18 (n=76 23.6%)	5 (n=13 38.5%)	0.533	-
Age	60.1 (9.8) (33.5 82.3)	66.0 (7.4) (48.0 84.9)	69.4 (7.4) (54.8 83.0)	< 0.001	a,b
Education	15.7 (2.87) (8.00 26.00)	15.0 (2.82) (5.00 22.00)	15.6 (4.91) (8.00 26.00)	0.0869	-
GDS	2.17 (2.40) (0.00 14.00)	2.86 (2.61) (0.00 13.00)	2.71 (2.11) (0.00 8.00)	0.058	-
UPSIT	23.41 (7.87) (5.00 39.00)	19.54 (8.70) (1.00 40.00)	16.06 (7.39) (5.00 31.00)	< 0.001	a,b
REM	3.82 (2.54) (0.00 12.00)	5.06 (2.90) (0.00 12.00)	5.24 (3.11) (0.00 12.00)	< 0.001	a
UPDRS3	20.51 (8.73) (5.00 47.00)	22.29 (9.40) (4.00 51.00)	21.29 (8.18) (10.00 39.00)	0.259	-
Cognitive outcomes					
MOCA	27.33 (2.17) (19.00 30.00)	26.58 (2.70) (17.00 30.00)	25.94 (2.38) (21.00 29.00)	0.003	
LNS	11.04 (2.50) (2.00 20.00)	9.29 (2.58) (4.00 17.00)	8.29 (2.78) (3.00 13.00)	< 0.001	a,b
SDMT	42.94 (8.75) (17.00 82.00)	36.69 (10.18) (7.00 58.00)	29.82 (10.85) (15.00 50.00)	< 0.001	a,b
JLO	12.98 (1.96) (5.00 15.00)	12.19 (2.44) (6.00 15.00)	11.53 (2.67) (6.00 15.00)	< 0.001	a,b
IMMEDIAT	25.64 (4.74) (9.00 36.00)	22.13 (4.90)(11.00 34.00)	20.24 (4.28) (14.00 28.00)	< 0.001	a,b
HVLTRDLY	8.75 (2.35) (0.00 12.00)	7.13 (2.75) (0.00 12.00)	7.00 (2.12) (3.00 11.00))	< 0.001	a,b
SFT	50.60 (11.54) (24.00 103.00)	43.48 (9.39) (20.00 70.00)	37.35 (9.37) (25.00 54.00))	< 0.001	a,b
ADL	93.25 (6.02) (70.00 100.00)	93.01 (5.29)(80.00 100.00)	91.76 (6.11) (80.00 100.00)	0.583	-
SCOPA	8.75 (5.70) (0.00 35.00)	11.89 (6.96) (3.00 39.00)	13.06 (7.20) (4.00 32.00)	< 0.001	a,b
CSF and DAT imaging markers					
$A\beta_{42}$	942.40 (413.12) (317.90 3707.00)	844.08(395.26)(238.80 2572.00)	592.27 (251.10) (308.80 1300.00)	0.001	b
ASYN	1519.17 (680.58) (432.40 5256.90)	1470.97(620.62)(472.00 3443.70)	1443.49 (642.42) (539.40 2780.50)	0.783	-
tTau	167.17 (54.61) (80.93 467.00)	172.53 (59.27) (83.93 345.30)	202.68 (78.07) (97.17 327.40)	0.053	-
pTau	14.52 (4.99) (8.02 40.13)	15.51 (5.40) (8.36 32.80)	18.59 (7.68) (8.01 30.84)	0.007	b
$pTau/A\beta_{42}$	0.0161 (0.0071) (0.0063 0.0673)	0.0196 (0.0120) (0.0103 0.0816)	0.0337 (0.0169) (0.0138 0.0698)	< 0.001	a,b
MEANPUT	0.85 (0.29) (0.24 2.15)	0.77 (0.32) (0.29 2.17)	0.66 (0.18) (0.31 1.05)	0.005	-
MEANCAU	2.07 (0.54) (0.39 3.71)	1.80 (0.54) (0.63 3.27)	1.59 (0.50) (0.74 2.52)	< 0.001	a,b

Table 2: Comparison among the results of the proposed predictive model for conversion to decline cognitive (using a binary classification, convert or not) with respect to the three highest risk factors (Age, Anosmia, A+) and MOCA at baseline.

Group	$Age > 65$	Anosmia	$A\beta_{42} < 683.5$	$MOCA < 25$	Model
PD-CU (n=323)	35.9%	28.2%	23.4%	20.1%	26.5%
PD-MCI (n=83)	53.0%	54.2%	26.5%	26.5%	63.9%
PDD (n=17)	76.5%	64.7%	66.7%	35.3%	94.7%

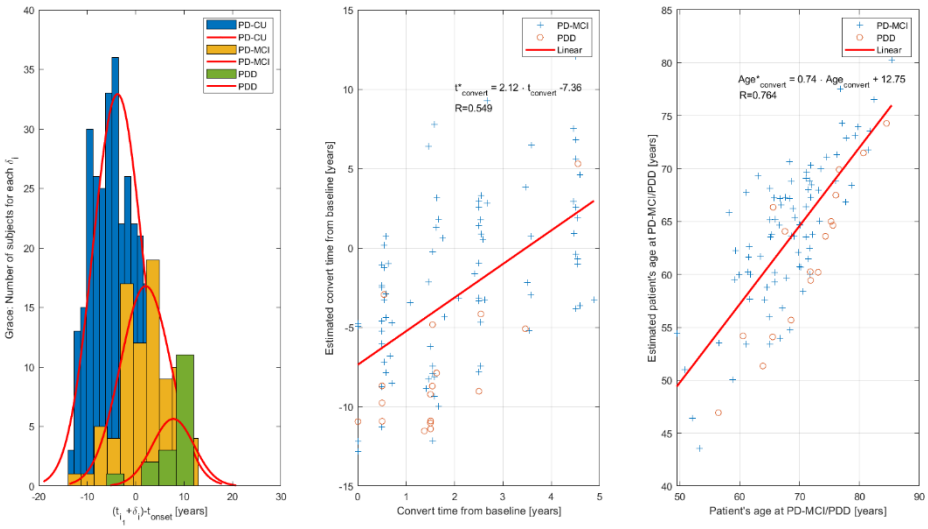


Figure 1: Classification of the subjects according to the time shift δ_i. The first column shows the histogram of the number of subjects as a function of the temporal ordering proposed by GRACE, $\left(t_{i_1}^c + \delta_i\right) - t_{onset}$. The second and third columns show the conversion times of the PD-MCI/PDD subjects with respect to their estimations by GRACE according to the cognitive reserve and patients' ages.

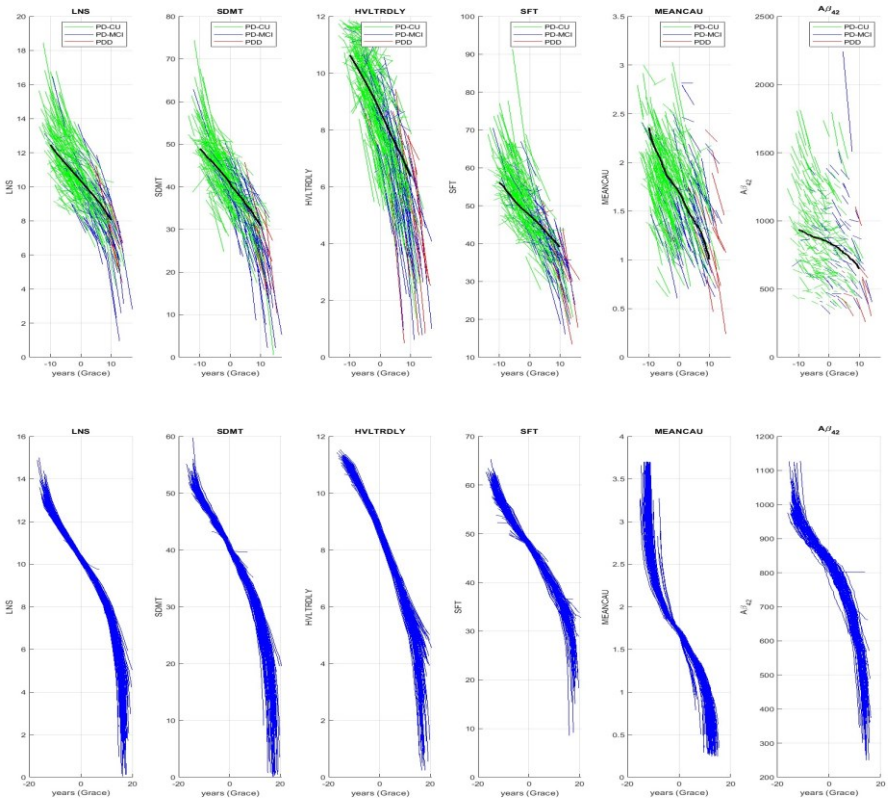


Figure 2: Temporary ordering proposed from PD-CU to PD-MCI/PDD by the selected DPM. In the first row, for each selected marker, the long-term trajectories (black lines) are superimposed over the subject-level observations in the original scale and colored according to diagnosis (PD-CU in green, PD-MCI in blue and PDD in red). In the second row, bootstrap analysis to obtain the confidence intervals of the long-term trajectories for each marker.

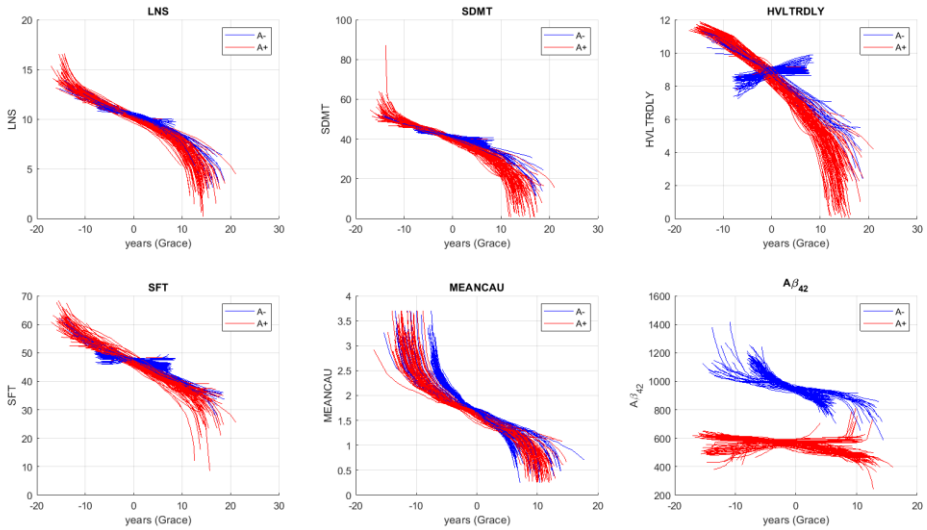


Figure 3: Long trajectories of the selected vector for discriminating between A+ and A- subjects. Use of the bootstrap technique to infer the confidence intervals of the long trajectories.

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